

EXPERIMENT No.10

OBJECT: - To study measure running and speed reversing the of a three phase Induction motor and record speed in both directions and calculation of slip.

APPARATUS REQUIRED:-

No.	Name of APPARATUS/IQUIPMENTS	TYPE	Specification /Range	Quantity
	Three phase variac	Induction	0-470v,15A	01
	Ammeter	MI	0-5A	01
	Voltmeter	MI	0-600V	01
	Three phase Induction motor	Induction	415V , 3.6A,1.5HP, 1440RPM, 50Hz.	01
	Tachometer	Digital	(0-9999) RPM	01
	Connecting wires			

THEORY: - Three phase Induction motor works on the principle of electromagnetic induction. It is an asynchronous motor. Rotor speed never reaches to synchronous value of speed. It always runs at speed slightly less than synchronous speed. Synchronous speed is calculated as:

$$\% \text{Slip} = \frac{(N_s - N_r)}{N_s} * 100$$

where N_r = Rotor speed in RPM.

Synchronous speed, $N_s = 120 f / P$ in RPM.

f = Frequency of supply.

P = No. of poles of motor

rotor always rotating magnetic field slips behind the speed known as synchronous speed.

speed can be reversed by inter changing any two phase of supply .

Observation Table:-

No.	V (Volt)	I (Amp.)	NS (RPM)	Nr (RPM)	%Slip	Mode
	415		1500			Running
	415		1500			Reversing

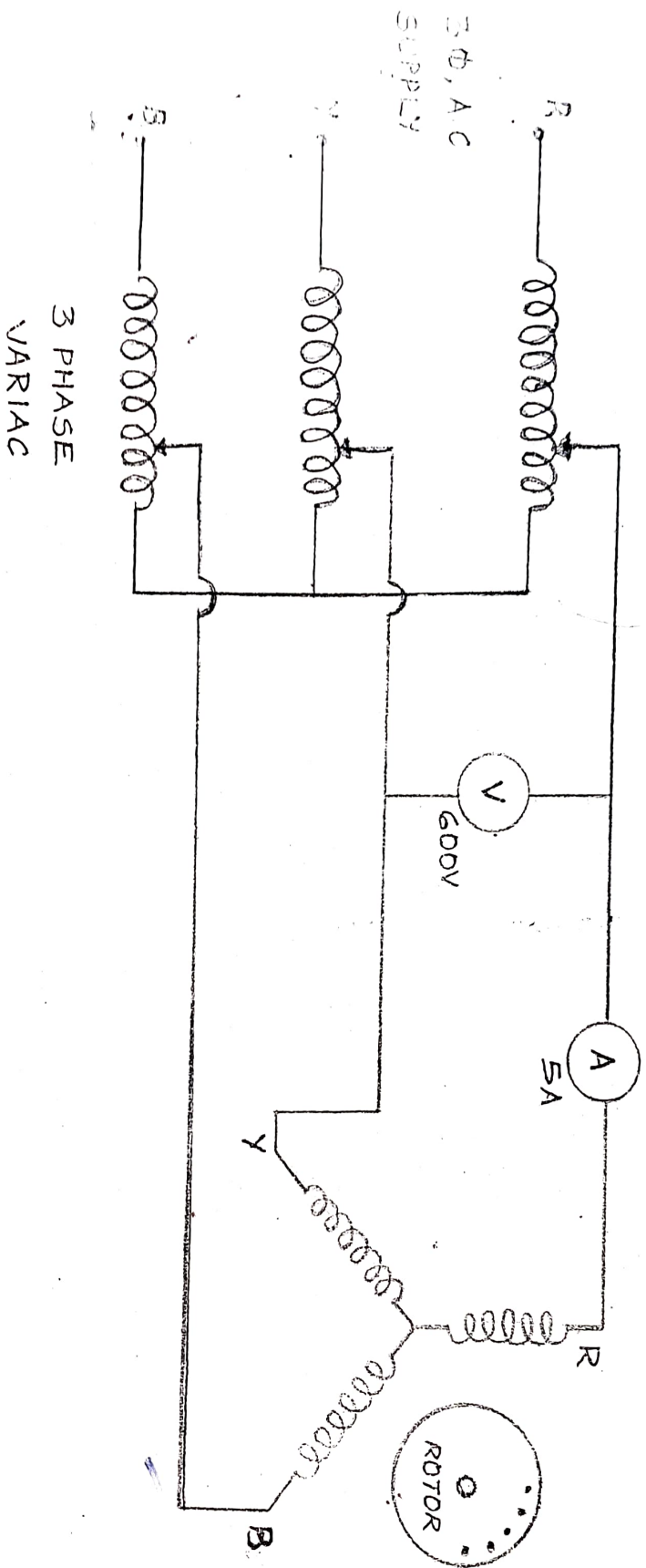
RESULT:- The running and speed reversal of a three phase induction motor has been studied & required measurement has been done.

PRECAUTION:-

- (1) All the connections should be tight and clear.
- (2) Start smoothly using three phase auto transformer.
- (3) Reading should be taken carefully with proper range of instruments.

REVIEW QUESTIONS:-

- Q1. What is the working principle of three phase induction motor?
- Q2. What are losses occurring in a three phase induction motor?



CIRCUIT DIAGRAM.